Bookstore System with Reader Club Features.

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1. Introduction

The Independent Bookstore System is a comprehensive web application built using PHP and MySQL. It provides customers with an intuitive platform to browse books, make purchases, and submit reviews, while empowering administrators with complete inventory and order management capabilities.

Core Objectives:

- Deliver seamless e-commerce experience for book lovers
- Enable post-purchase review system with star ratings
- Provide administrators with robust management tools
- Support reader club features through purchase verification

2. Functional Requirements

GUEST USERS

FR-01	Register new account with email verification
FR-02	Browse books by title, author, or genre
FR-03	Search books with real-time filtering
FR-04	View book details (price, description, ratings)

REGISTERED USERS

FR-05	Secure login/logout with session management
FR-06	Add books to persistent shopping cart
FR-07	Manage cart (update quantities, remove items)
FR-08	Place orders with shipping address input
FR-09	View order history and status tracking
FR-10	Write reviews (1-5 stars) for purchased books only

READER CLUB FEATURES

FR-11	Create and join book clubs (public/private)
FR-12	Collaborative reading schedule with milestone tracking
FR-13	Vote for next book using weighted polling system
FR-14	Member spotlight algorithm (Reader of the Month)
FR-15	Reading challenge manager (e.g., 52 books/year)

ADMINISTRATORS

FR-16	Admin dashboard with role-based access control
FR-17	CRUD operations for book inventory management
FR-18	View and update all customer orders
FR-19	Manage user accounts (activate/deactivate, promote)
FR-20	Generate basic reports (top reviewed books, sales)

SYSTEM BUSINESS RULES

FR-21	Stock validation before order confirmation
FR-22	Automatic average rating calculation
FR-23	Purchase verification for review eligibility
FR-24	Multi-vendor cart splitter for orders from multiple stores
FR-25	Cross-store inventory sync engine (real-time)

3.(non) Functional Requirements

PERFORMANCE

NFR-01	Pages load within 3 seconds under normal load
NFR-02	System supports up to 100 concurrent users without degradation
NFR-03	Search results return in less than 1 second
NFR-04	Database queries execute in under 200ms

SECURITY

NFR-05	Passwords hashed using bcrypt (cost factor = 10)
NFR-06	Prepared statements for all database queries (SQL injection prevention)
NFR-07	Session tokens expire after 24 hours of inactivity
NFR-08	Role-based access control enforced at API level

USABILITY

NFR-09	Fully responsive design (mobile, tablet, desktop)
NFR-10	WCAG 2.1 AA compliance for accessibility
NFR-11	Clear error messages with suggested actions

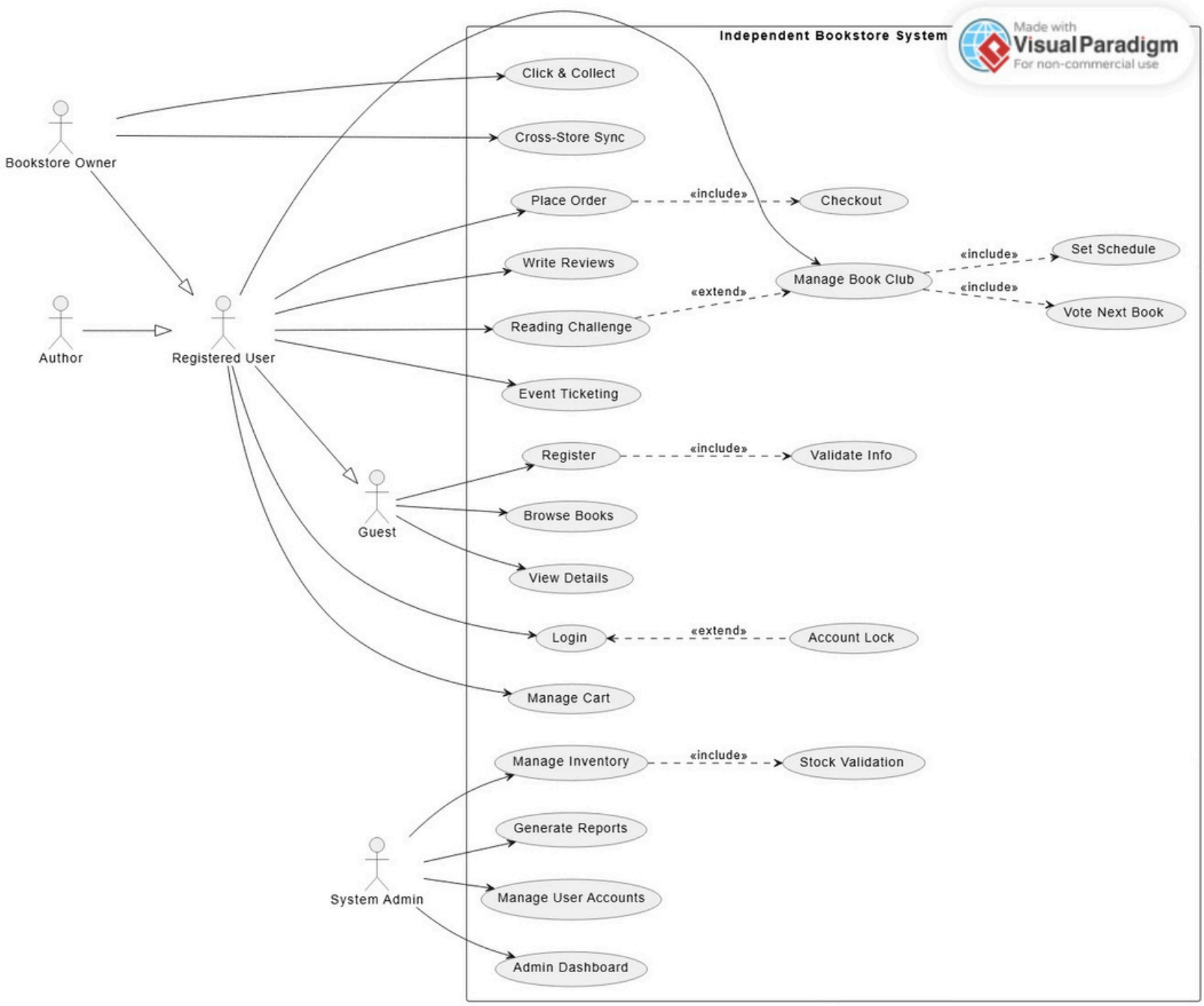
RELIABILITY & SCALABILITY

NFR-12	99.5% uptime during business hours
NFR-13	Transaction rollback on any stock validation failure
NFR-14	Modular architecture allowing horizontal scaling

DESIGN & IMPLEMENTATION CONSTRAINTS

C-01	Backend: PHP 8+ with secure coding practices
C-02	Database: MySQL 8+ with optimized queries
C-03	Frontend: HTML5/CSS3/JavaScript (vanilla, no heavy frameworks)
C-04	Local development: XAMPP/WAMP
C-05	Production-ready code structure with easy cloud migration

4.USE CASE Daigram



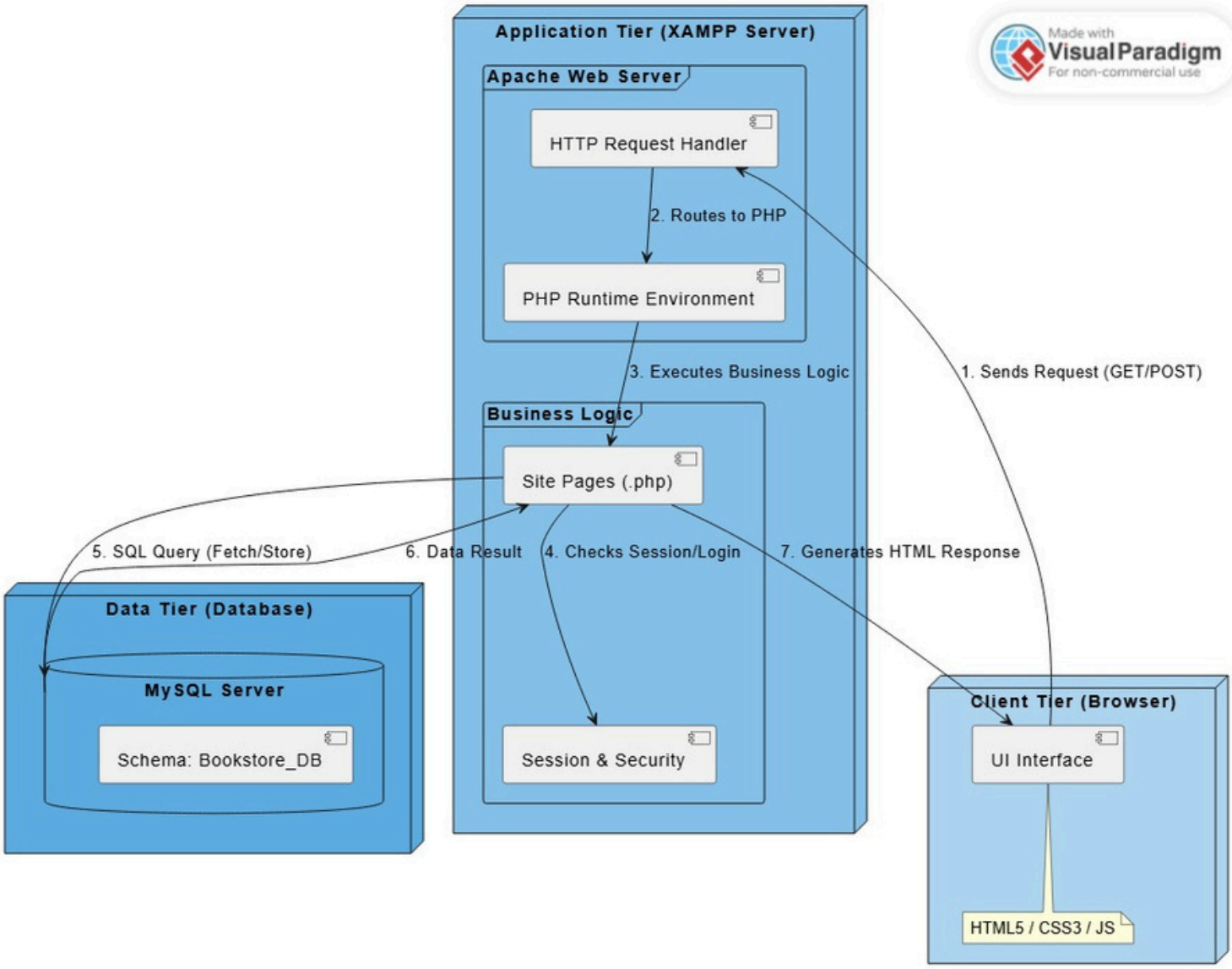
4.1 use case Description:

This diagram shows the main use cases of the system and how users interact with them.

It gives a general overview of the system functionalities.

Detailed use case diagrams are also provided separately.

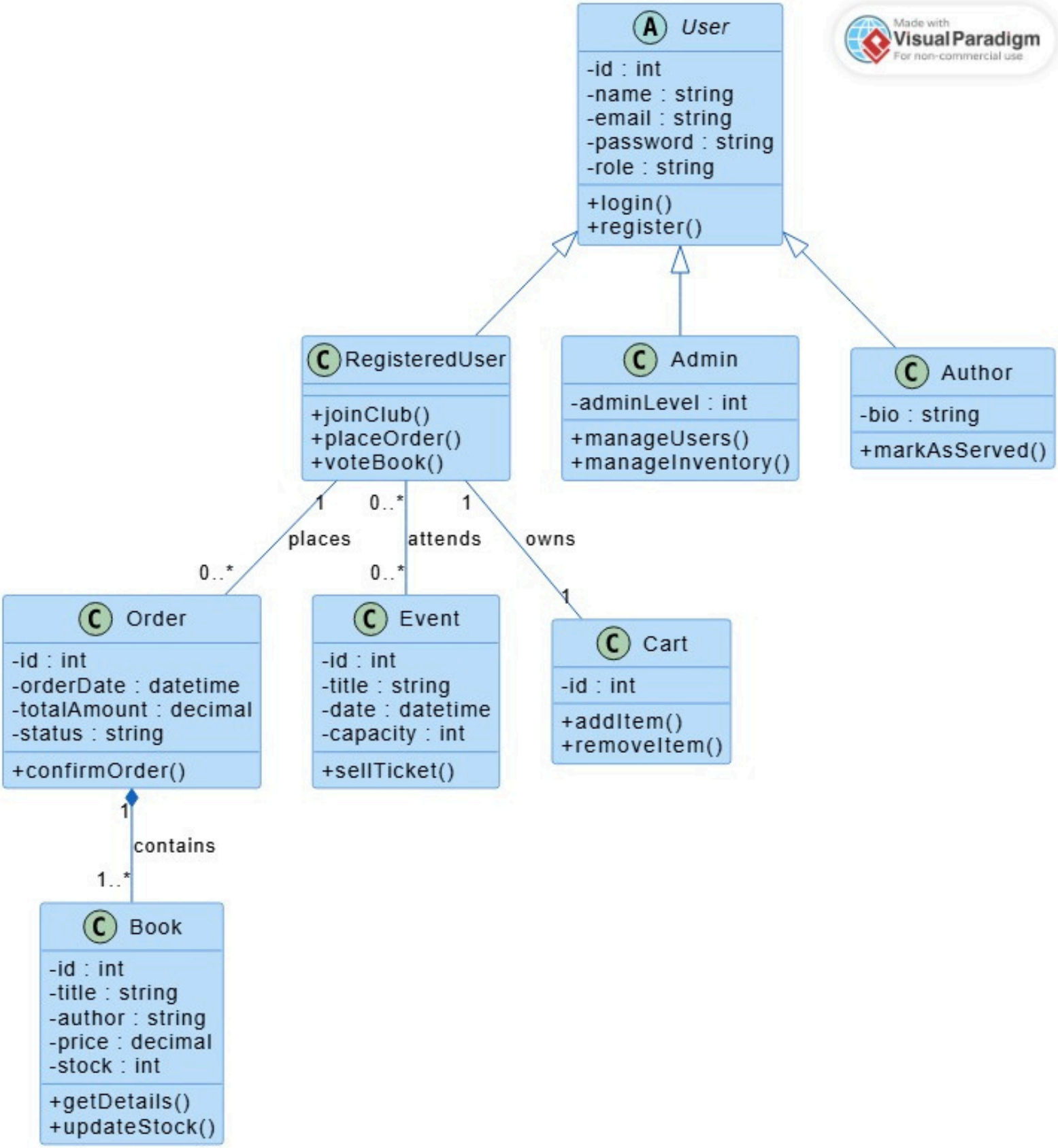
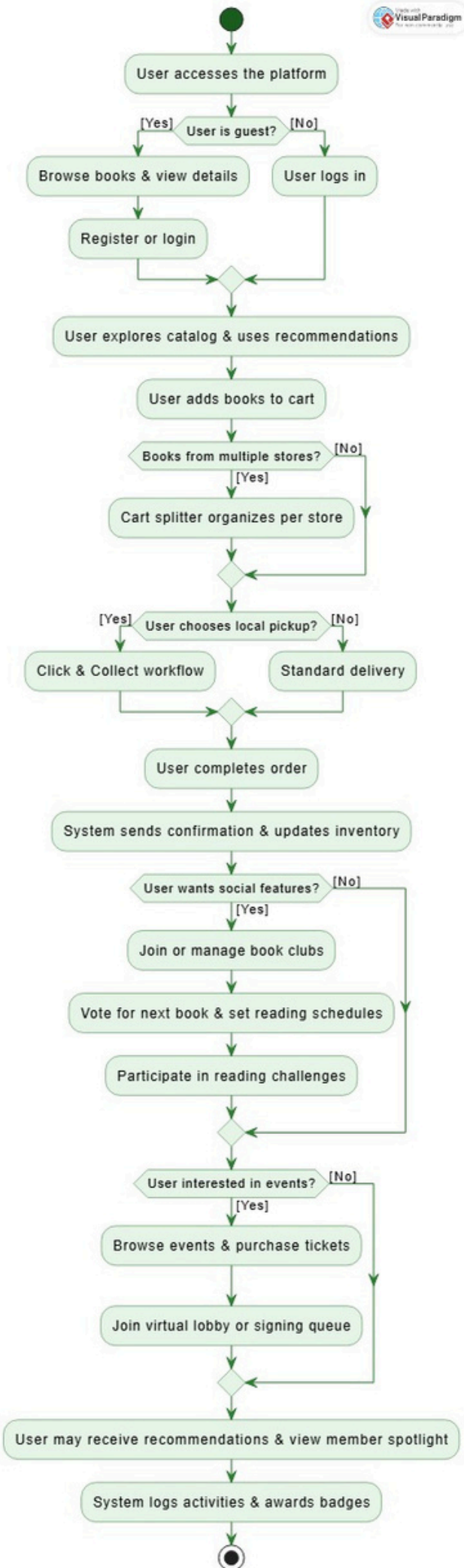
5. System Architecture:



5.1 System Architecture details:

A Start version of our System Architechure shows tech-stack and a pre-vision for how the system will Be.

6.CLASS Diagram



7. *ACTIVITY Diagram*